UNIVERSITY OF LEICESTER

SCHOOL OF ENGINEERING

EG7010 – ENGINEERING DESIGN CASE STUDY

Gear Train Assembly

SolidWorks Assignment

Use SolidWorks to Model a Gear Train Assembly

Use SolidWorks to create a gear train assembly by modelling Spur Gears of your own choice from the range available from the on-line catalogue of https://www.hpcgears.com/. Figure 1 - 4:1 Gear Train Modelled using SolidWorks, shows the type of Gear Train you could produce.

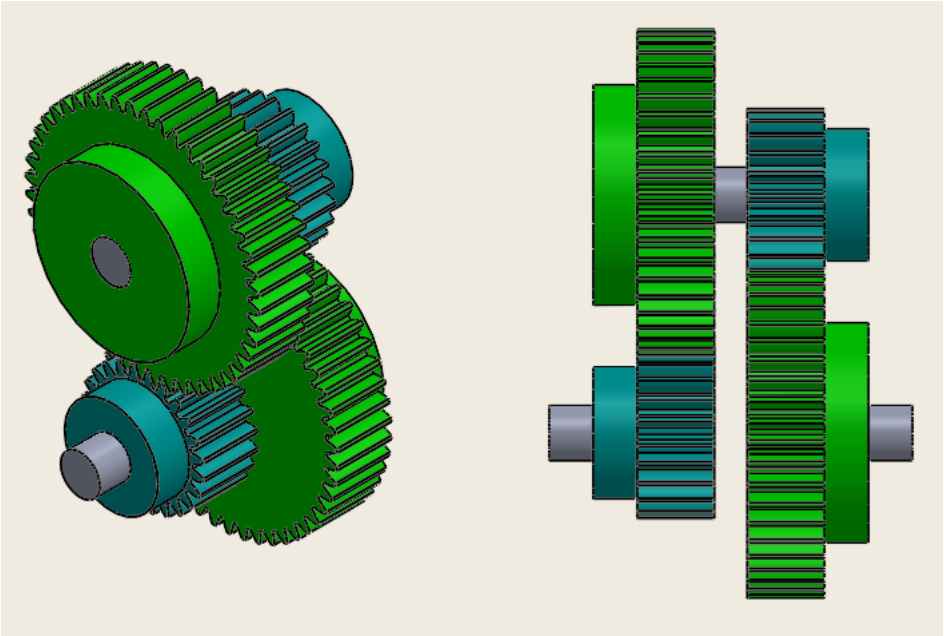


Figure 1 - 4:1 Gear Train Modelled using SolidWorks

You must consider and complete the following.

1. The gear train must consist of at least three spur gears, but you can use the same models more than once within the assembly. You are not allowed to use YG2.5–28 as part of your gear train.
2. The Input/Out speed ratio must be at least 1:4 or 4:1 depending on whether you choose a speed increasing or decreasing design.
3. Create SolidWorks Part Model of your chosen Spur Gears using 'Unwin's Construction Method'. The part files must be original and must not be a configuration taken from the SolidWorks Toolbox or any online repository such as GrabCAD. Files created before Wednesday 30th December 2022 will not be considered for marking.
4. Create any additional shaft models you feel the assembly requires.
5. Create a SolidWorks Assembly Model in which you combine the Spur Gears and any other parts you create

6. Using the drawing file Gear YG2.5-28 as a reference available here;

   Reference Spur Gear YG2.5-28 Drawing

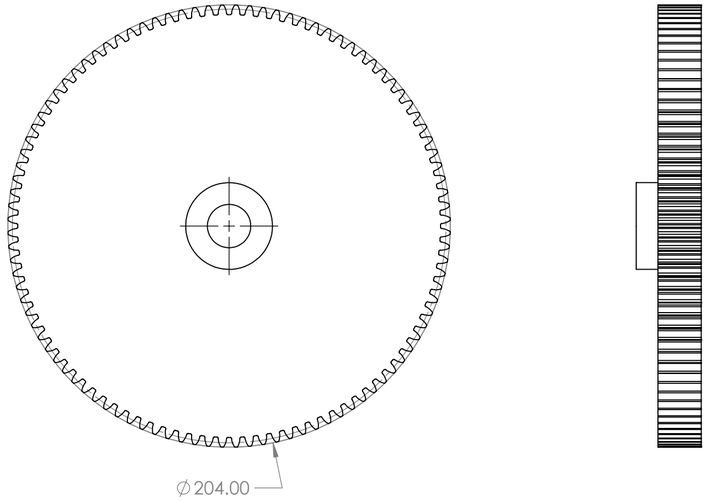
   Use a University of Leicester drawing template to create a SolidWorks drawing which can be used to quickly reproduce a SolidWorks part model of one of your Spur Gears without the need to use 'Unwin's Construction Method'.

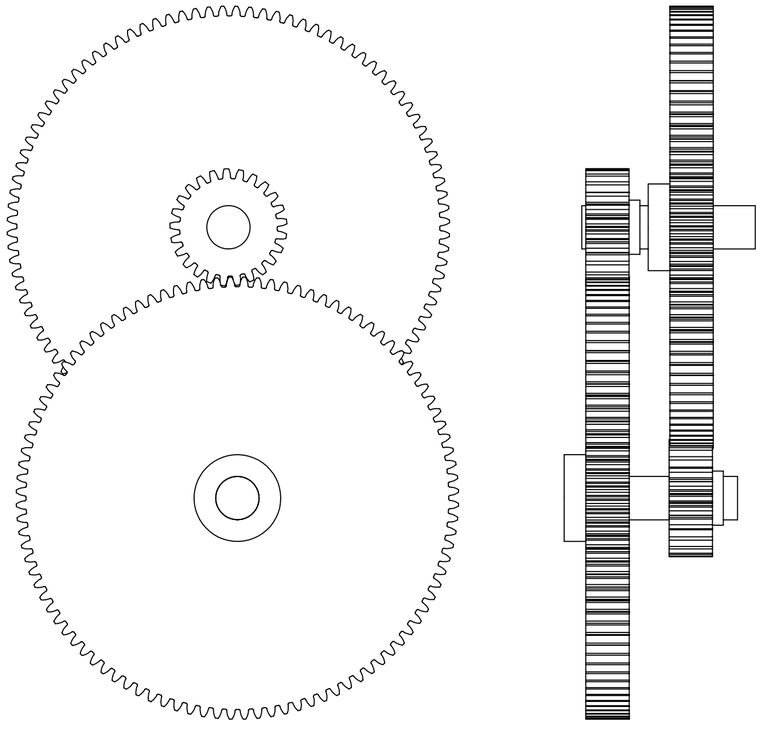
7. Use a University of Leicester drawing template to create a balloon referenced assembly drawing of your assembly which should include a parts list.

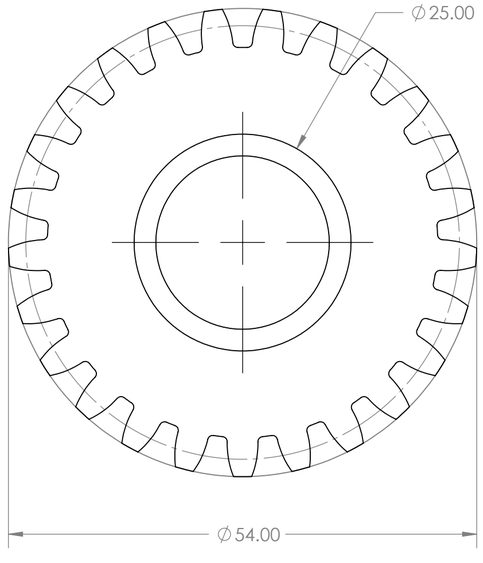
You are required to upload a compressed folder containing all of the SolidWorks models, assemblies and drawing files you created in order to complete the assignment.

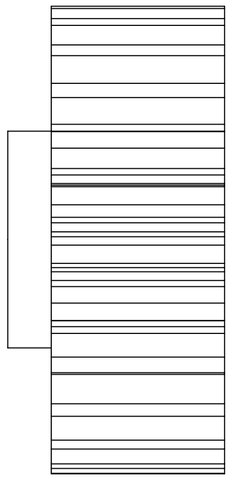
Only SolidWorks files will be considered for marking, files submitted in other formats such as .PDF will not be marked.

The deadline for your submission is 17:00 Friday 16th December 2022.

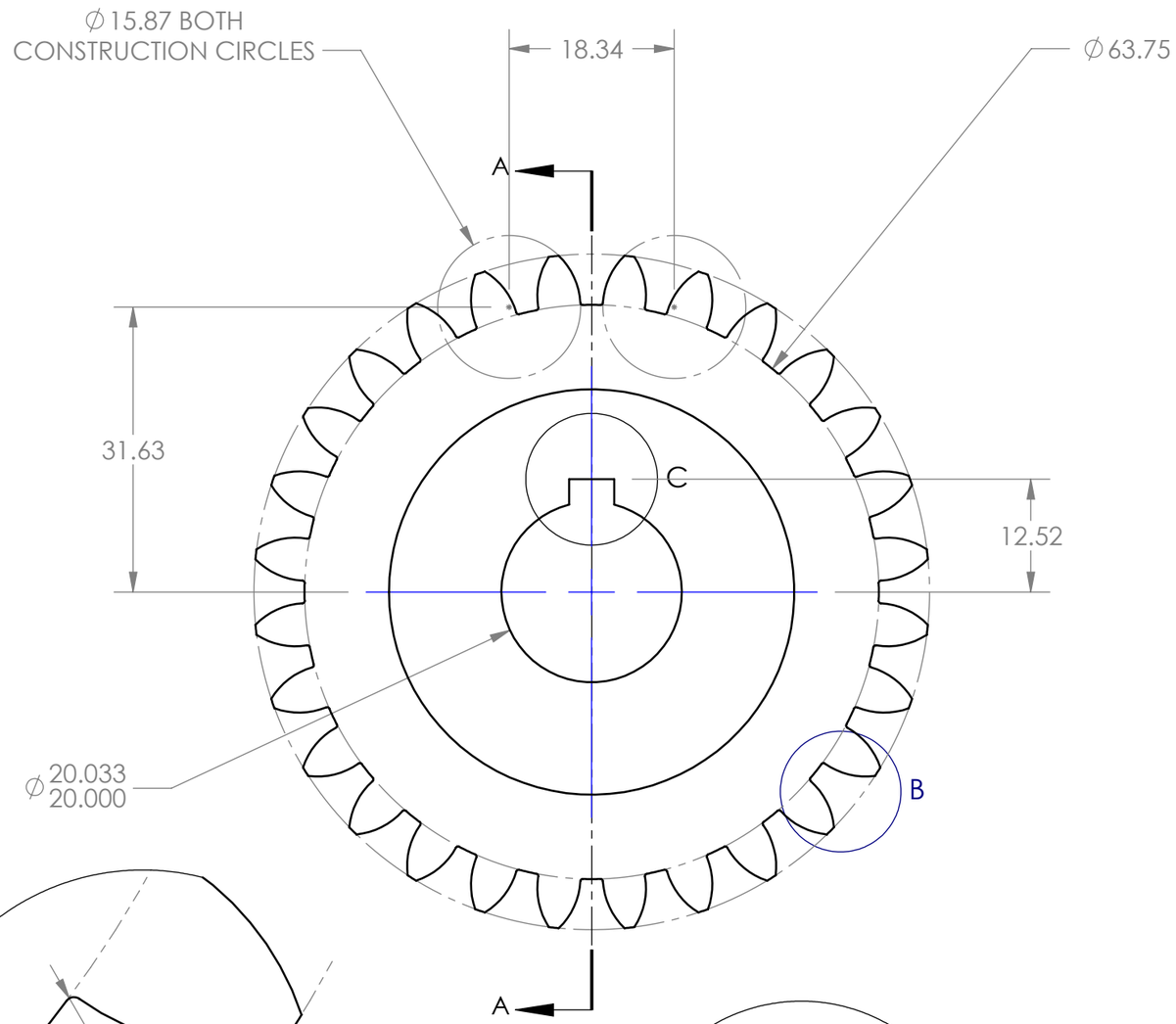


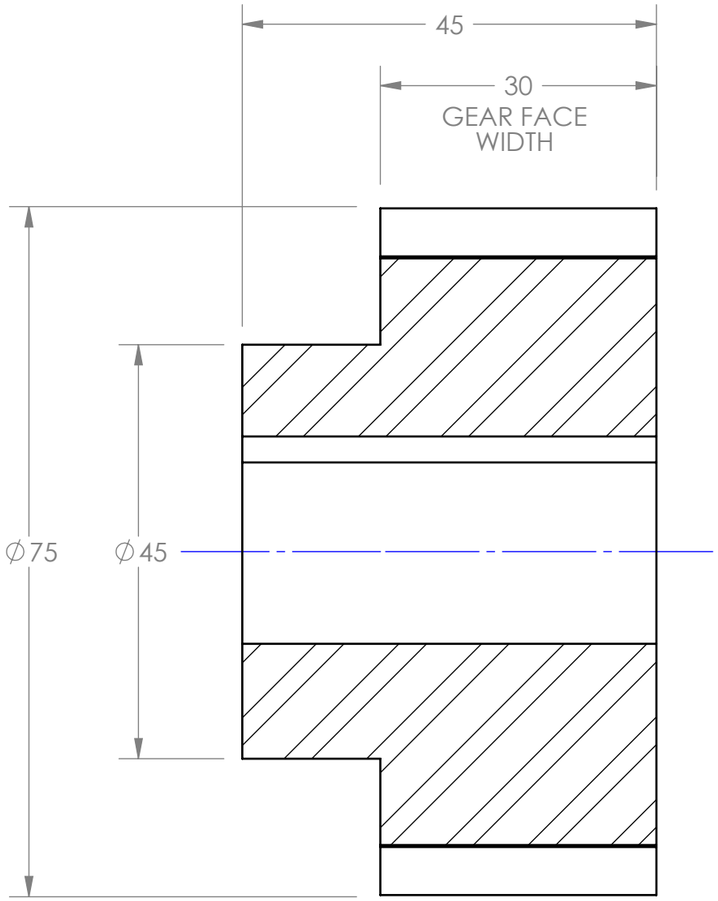




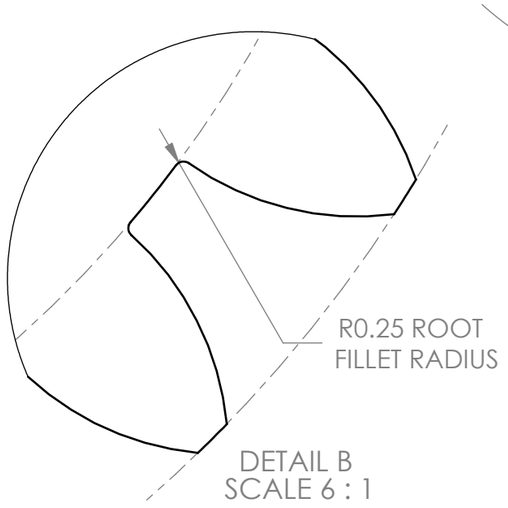


UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: LINEAR: ANGULAR:			FINISH: DEBURR AND BREAK SHARP EDGES	UNIVERSITY OF LEICESTER		DO NOT SCALE DRAWING / REVISION
				SCHOOL OF ENGINEERING		
	NAME	SIGNATURE	DATE			TITLE:
DRAWN						
CHK'D						
APPV'D						
MFG						
Q.A				MATERIAL:		DWG NO. GEAR ASM / A2
				WEIGHT:		SCALE:1:2 / SHEET 1 OF 1

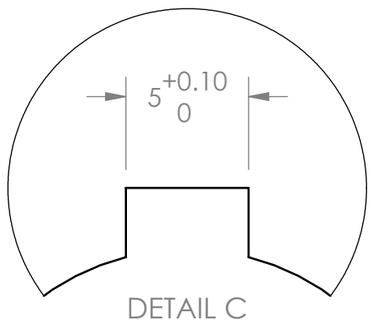




SECTION A-A
SCALE 1.5 : 1



DETAIL B
SCALE 6 : 1



DETAIL C
SCALE 4 : 1

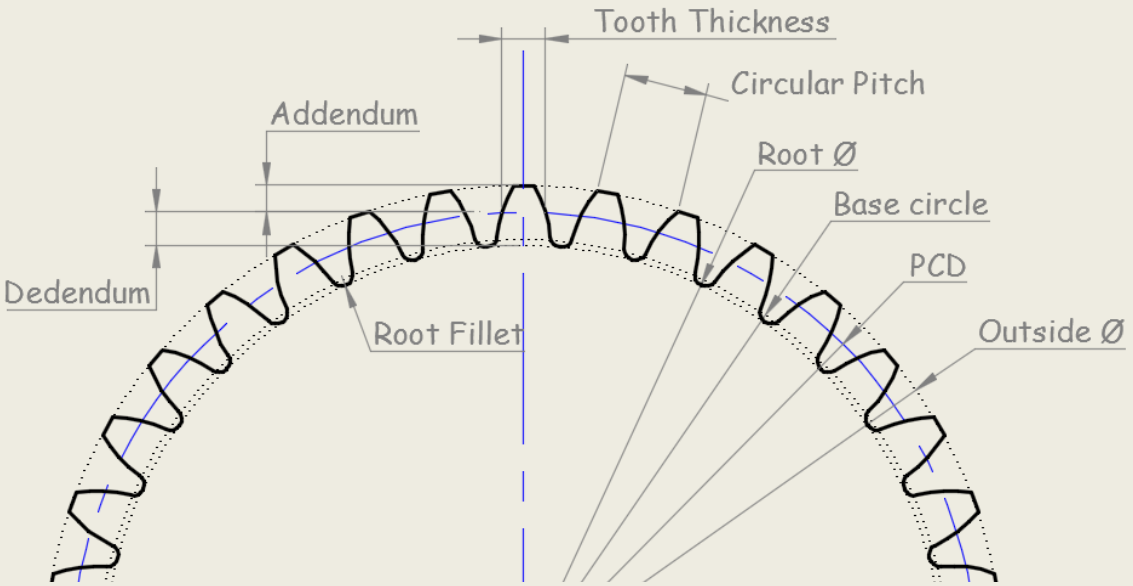
UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: LINEAR: ANGULAR:	FINISH: DEBURR AND BREAK SHARP EDGES	UNIVERSITY OF LEICESTER	DO NOT SCALE DRAWING / REVISION
			SCHOOL OF ENGINEERING
	NAME / SIGNATURE / DATE		TITLE: UNWINS GEAR REPRESENTATION
DRAWN			
CHK'D			
APPV'D			
MFG			
Q.A		MATERIAL:	DWG NO. YG2.5-28 / A3
WEIGHT:		SCALE:1:1	SHEET 1 OF 1

Gear Construction Method

The method presented here to draw a typical spur gear is called 'Unwin's Construction Method'; it will give an approximate gear figure and not a true involute curve.

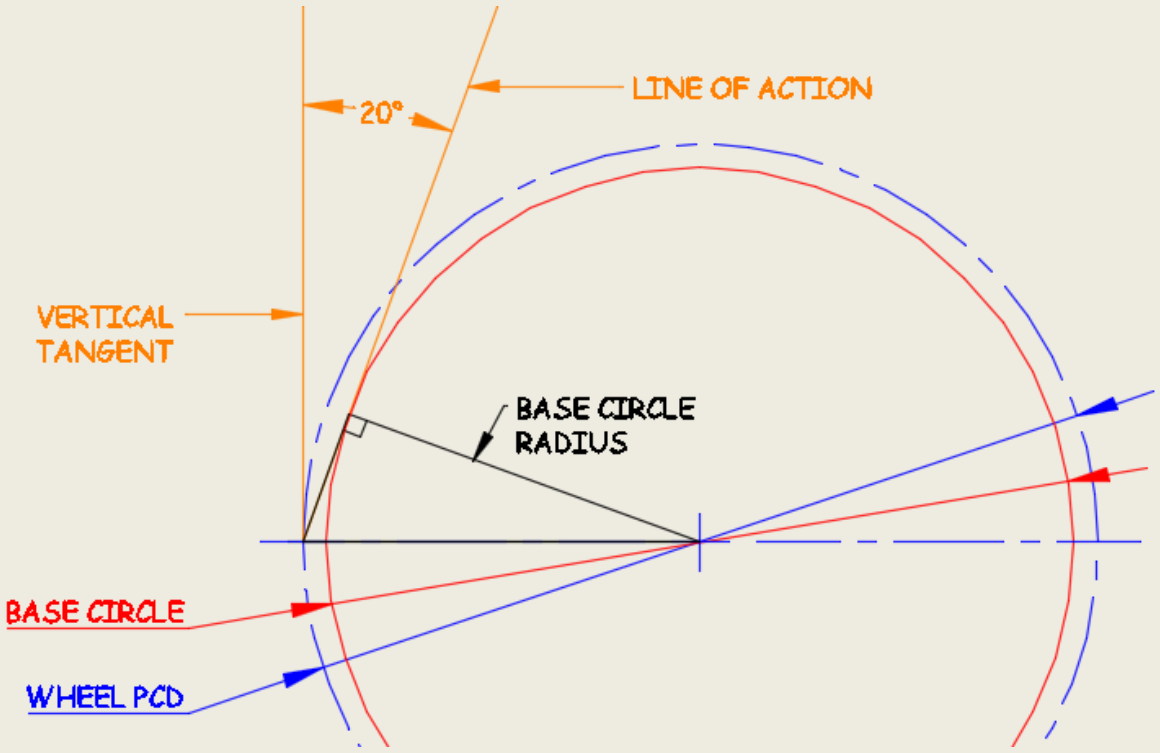
You will need the following information about the gear in order to construct it;

- Pressure angle – Normally 20°
- Gear Module - The unit of size which represents how big the gear teeth are.
- Pitch Circle Diameter (PCD) – Module x Number of Teeth. The PCD can be considered as the mating diameters between two gears.
- Addendum – The part of the tooth above the PCD equal to the Gear Module
- Dedendum – The part of the tooth below the PCD equal to 1.25 x the Gear Module
- Circular Pitch – The distance from a point on one tooth to the same point on the next tooth equal to the Gear Module x π
- Tooth Thickness equal to the Circular Pitch / 2
- Root Fillet Radius (an arbitrary size of your choice).

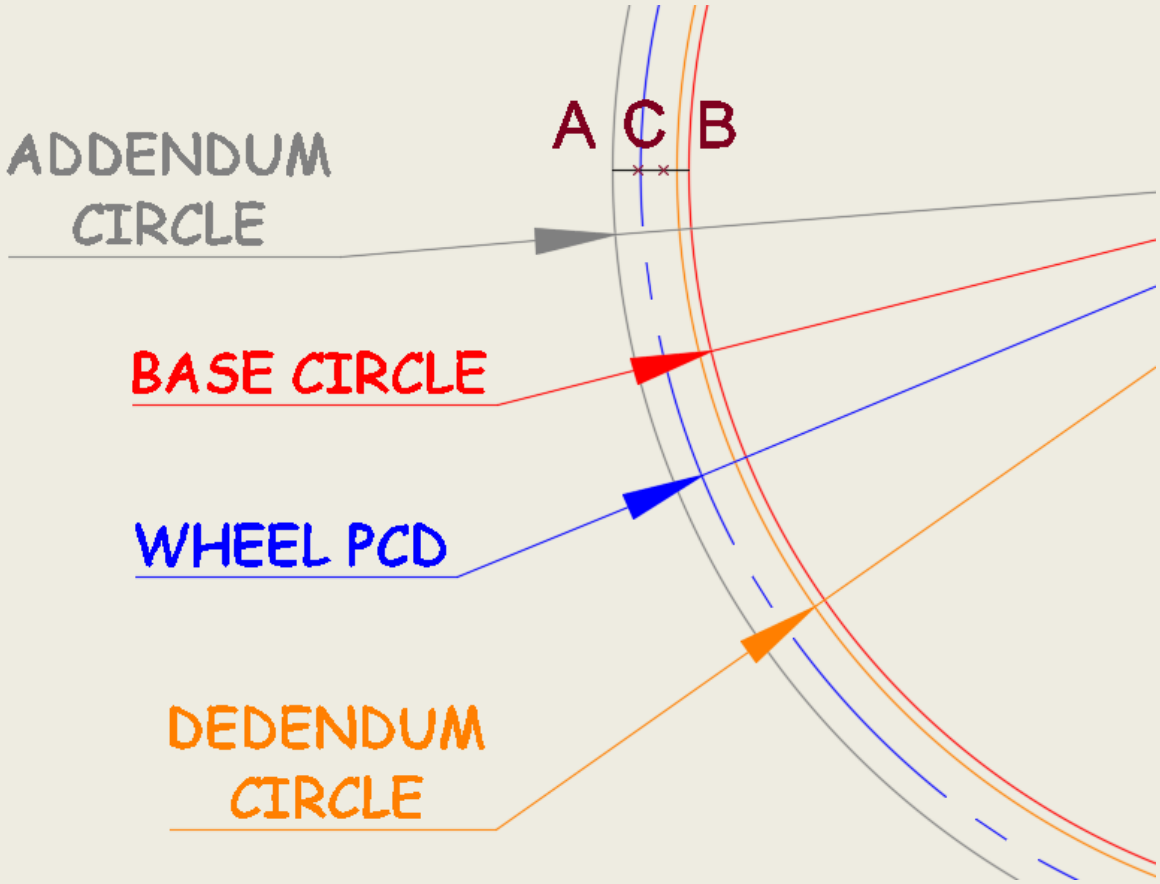


Unwin's Gear Construction Method

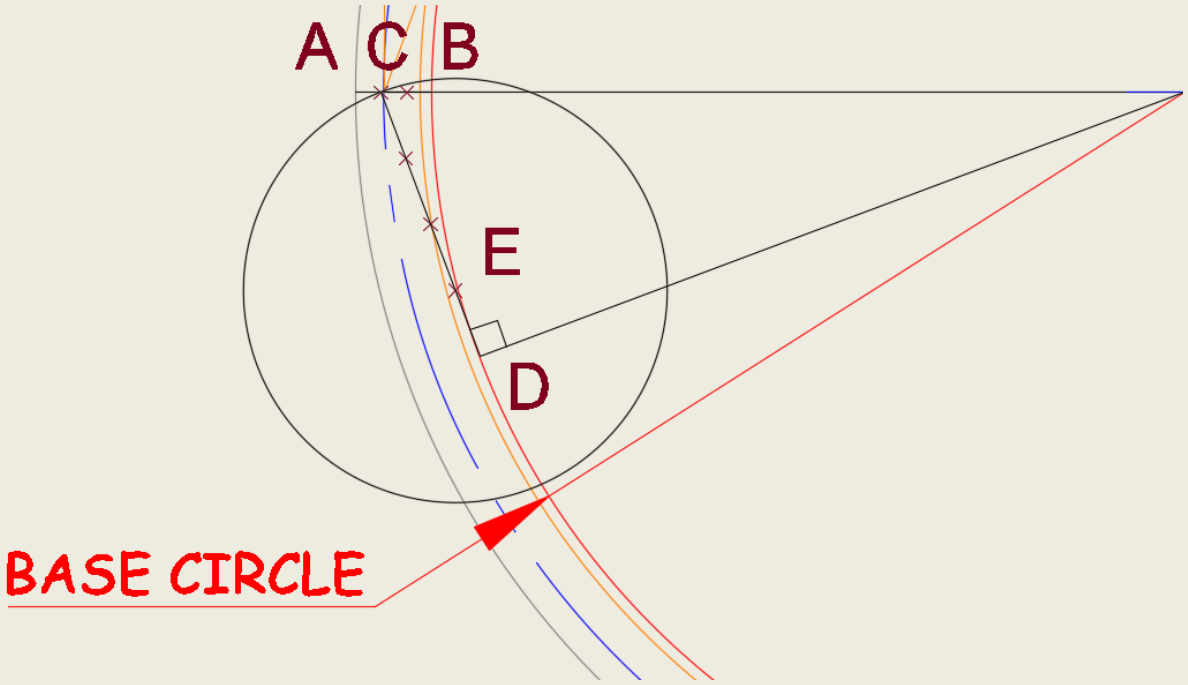
1. Calculate all required dimensions before you begin to sketch or use the equation editor within SolidWorks.
2. Open a new sketch, draw the pitch circle (labelled WHEEL PCD in the figure below). The geometry in the figures has been coloured for clarity. You will not be able to change the colour of the SolidWorks sketch geometry.
3. Draw a vertical tangent line from a quadrant of the pitch circle.
4. Draw a line at 20° to the vertical tangent to denote the line of action
5. Draw a line from the centre of the pitch circle perpendicular to the line of action.
6. This line forms the radius of the base circle, draw the base circle.



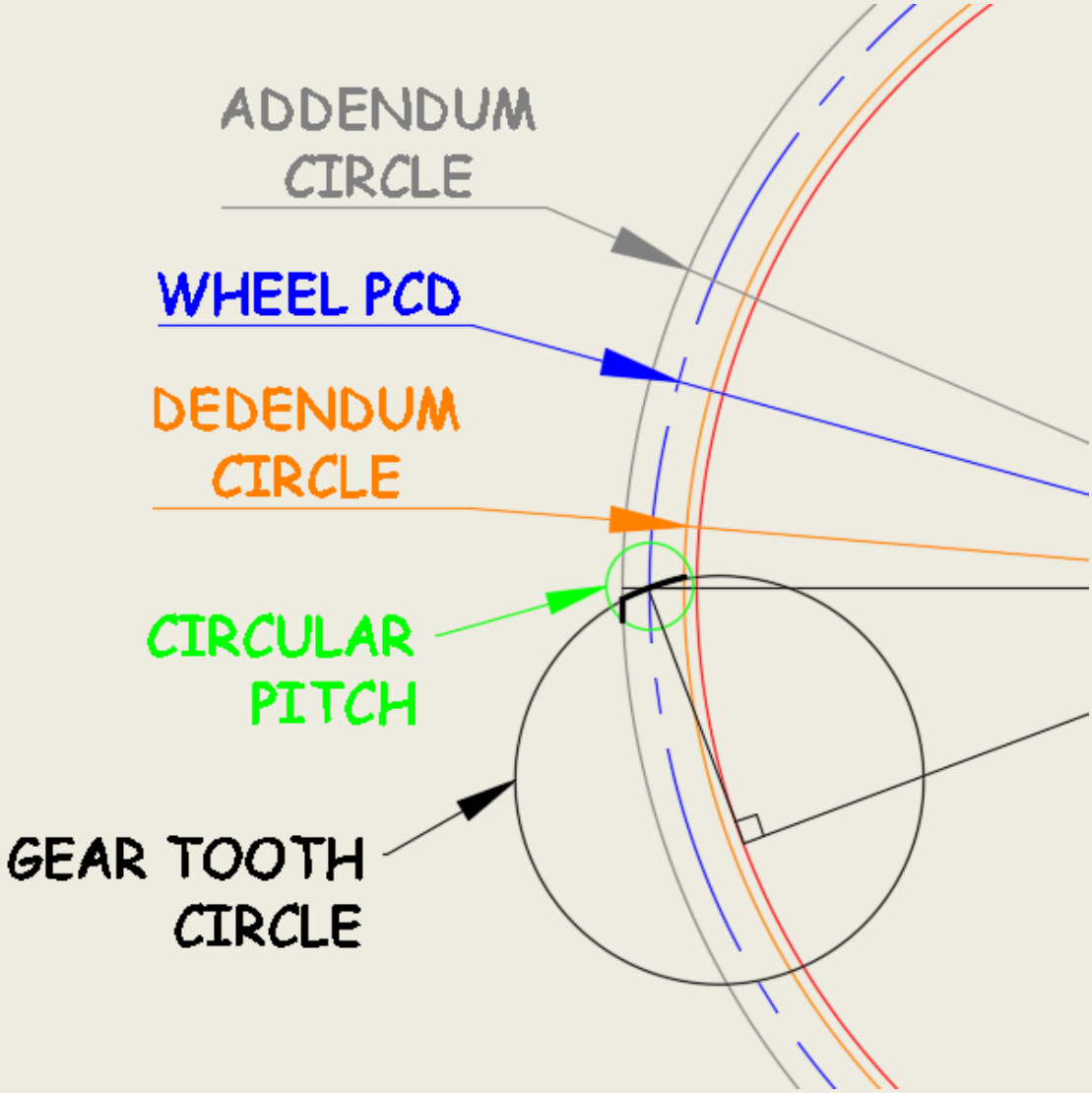
7. Draw the addendum and dedendum circles. The dedendum circle could be either inside or outside of the base circle depending on the size of the gear you are constructing.
8. Draw line AB between the quadrant of the base circle and the quadrant of the addendum circle.
9. Position point C so that AC = AB/3. The equations tool can be used to accurately position a point on a line.



10. From point C, draw a line CD tangent to the base circle.
11. Position point E so that ED = CD/4. The equations tool can be used to accurately position a point on a line.
12. Draw the gear tooth circle with centre E and radius CE.

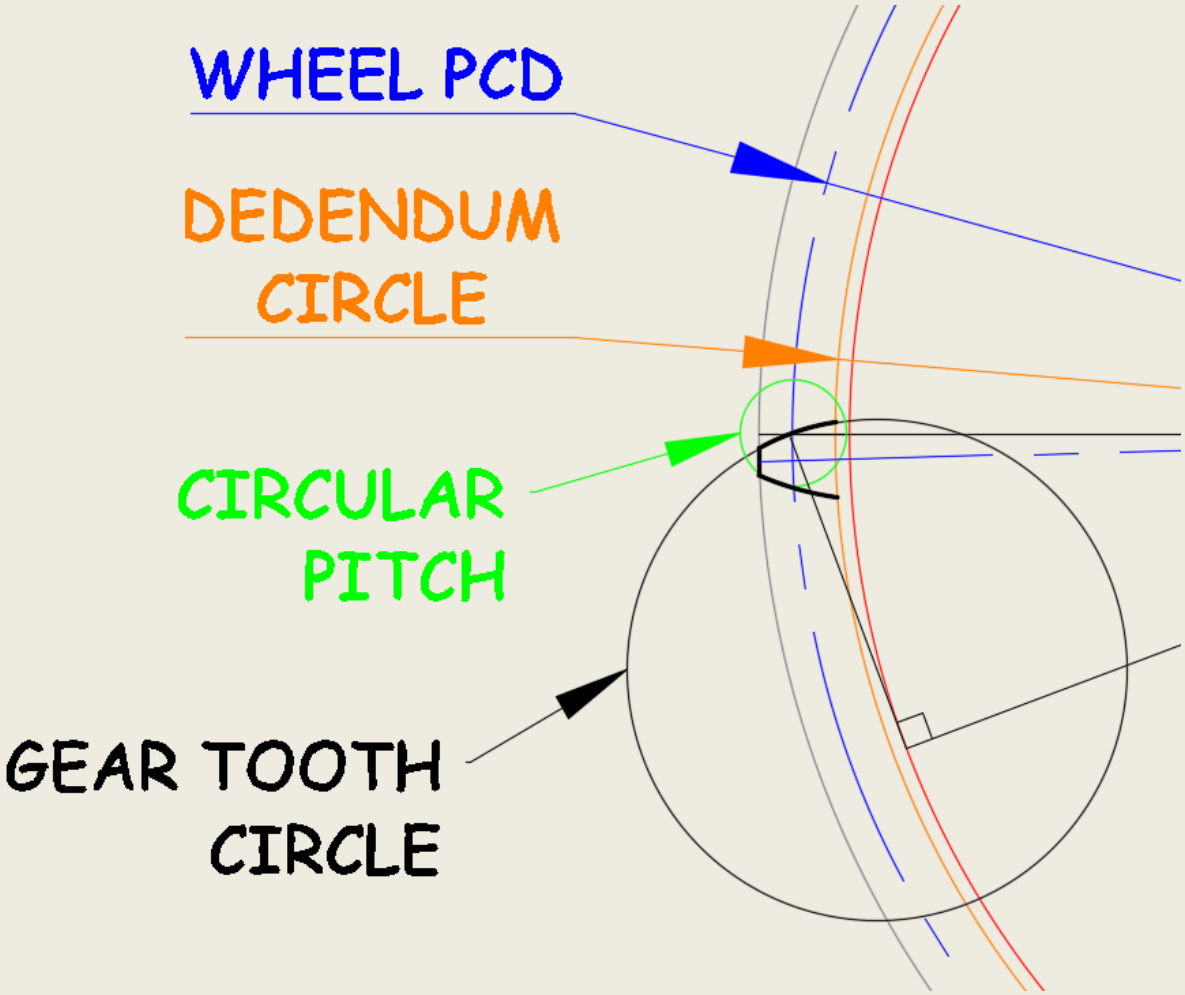


13. Draw a circle of diameter equal to the circular pitch with its centre at the intersection between the GEAR TOOTH CIRCLE and the WHEEL PCD. You will need to use sketch relations to complete this step.
14. The arcs highlighted between the gear tooth and circular pitch circles are outside edge of the tooth. Note the gear tooth arc extends to touch the dedendum circle.

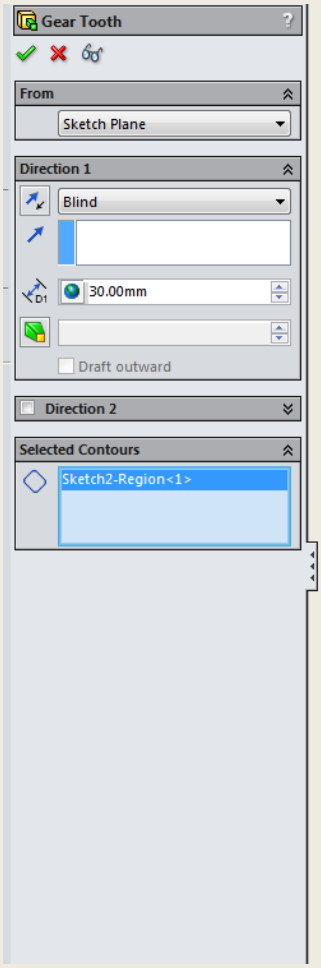


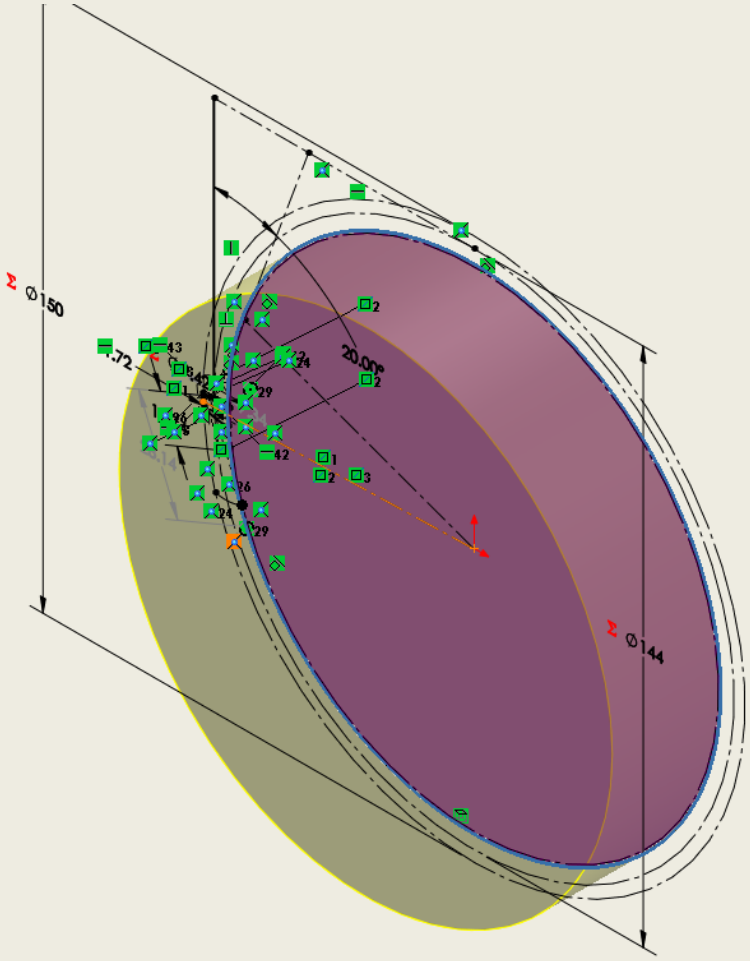
15. Either trim the gear tooth and addendum circles or draw arcs to form the tooth.

16. Complete the gear tooth using the MIRROR command.

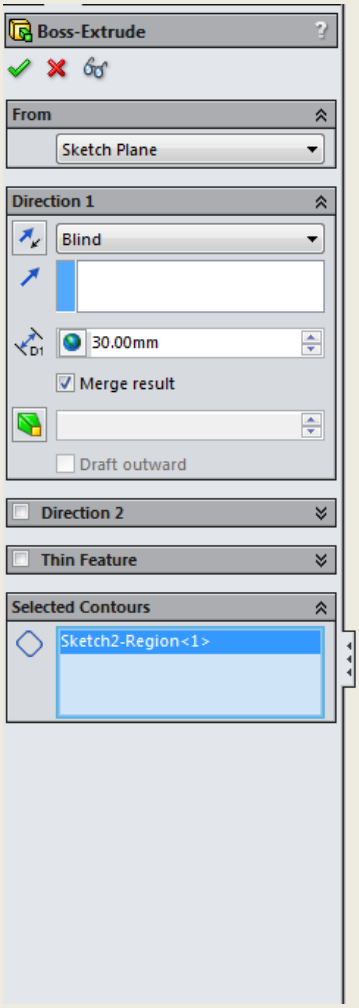


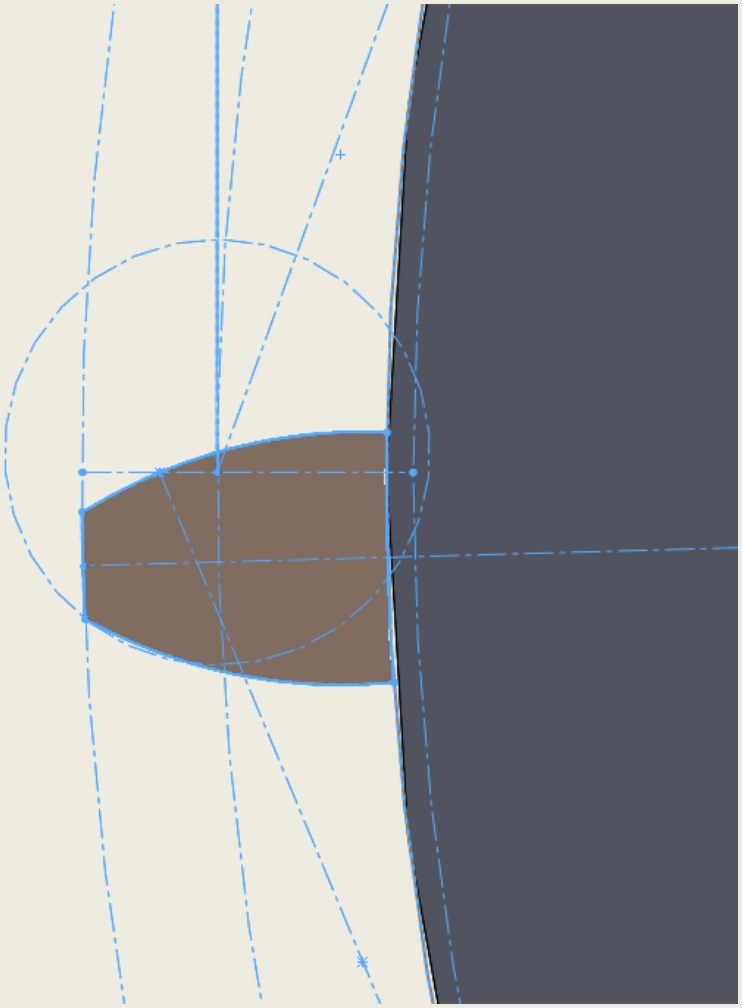
17. Save and Exit the sketch and select Extruded Boss/Base.
18. Choose the existing sketch to Extrude and specify the extrusion thickness.
19. In the Selected Contours box choose the area inside the dedendum circle just to extrude the circle and select OK.



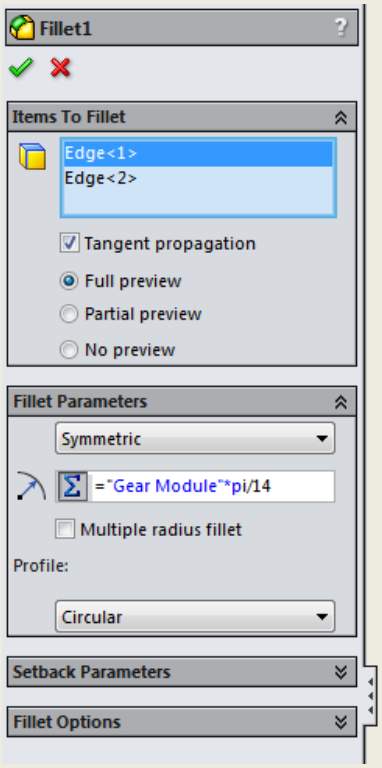


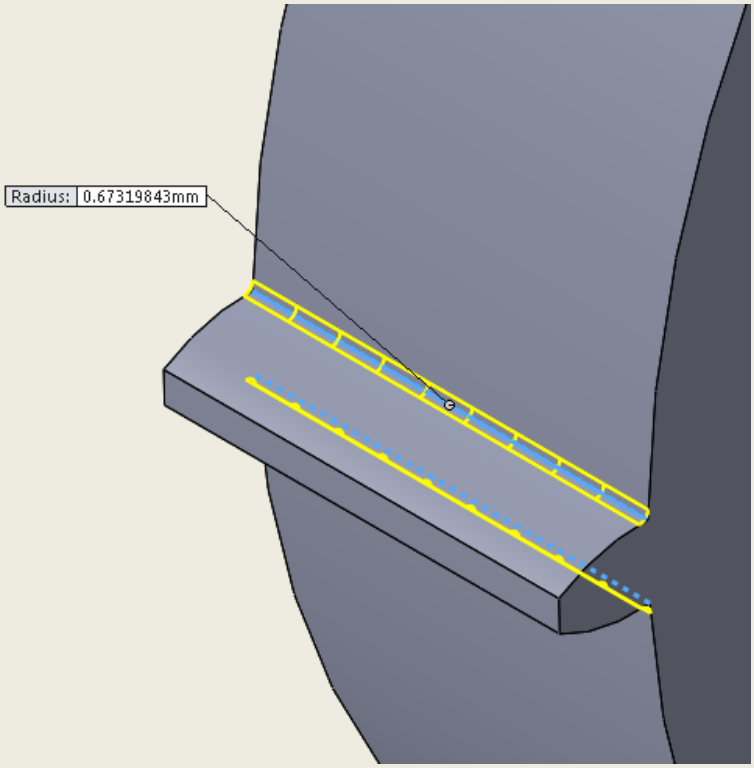
20. Repeat the extrusion process using the same sketch but this time select the area inside the gear tooth as the Selected Contour. This will extrude one tooth onto the gear.



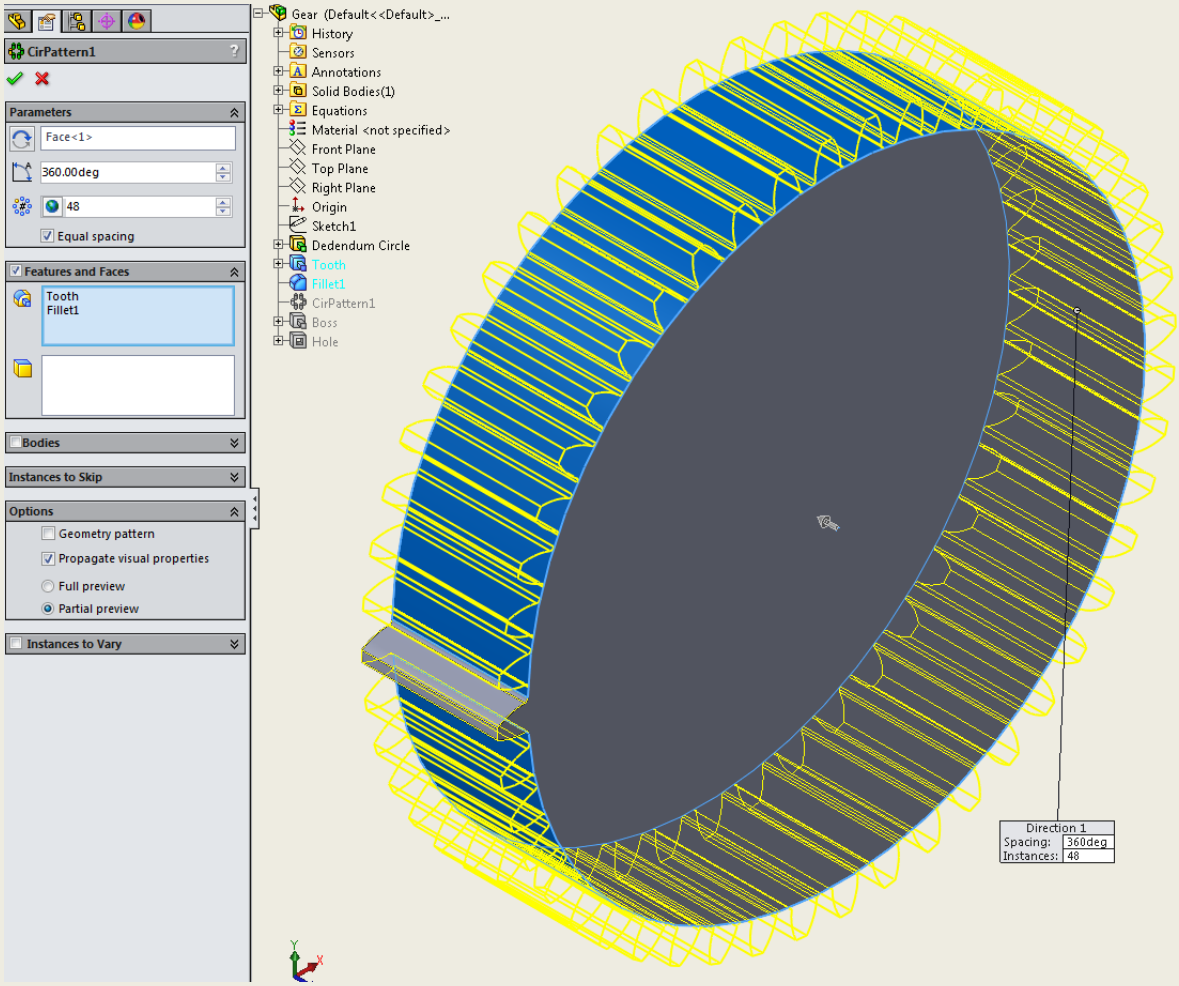


21. Add the root fillets to the gear tooth.





22. Array the Tooth and Fillets using the Circular pattern tool.



23. Finish the gear by adding the boss extrusion and the bore extruded cut.

